Sibling points Stage 2: prime-field shapes

Sibling points, Stage 2: prime-field curve shapes (results, local L0)

What ran. The frozen panel of `PROTOCOL.md` (commit `2faf4ecaf9`, frozen before data). It covers three prime-field shapes, each with five prime-order curves (31-bit p, r from 2^30.1 to 2^31): - `a3`: a = -3, the NIST P-curve shape. - `gen`: random a and b. - `j0b7`: j = 0, b = 7, the secp256k1 shape.

Each curve has 8 public targets, run for 6 rounds plus 1 warmup. The arms are rho-neg, rho-neg-aa (A/A), tw2cj-lazy, sib8-p64, sib8-p256 and sib8-p512, plus rho-cubic (A = 6) on j0b7. The panel ran on this session's 4-vCPU VM at L0, under bench record `benchmarks/records/20261111T0020Z-sib-stage2-primefield.json`.

The first launch hit "no space left on device" while a worktree was being created. That partial session was deleted and the same frozen specs were re-run unchanged.

Admission. All three specs pass: 1,680 + 1,680 + 1,960 records verified, `verify --replay 8` reproduced 8 of 8 per spec with 0 problems, and the A/A ratio is exactly 1 in the counted unit. All numbers are `[pcc:unverified]`, with no pcc claim filed and no validation gate.

Results: paired ratios over 240 pairs on 40 workloads and 5 curves (`ecbench compare`, `results.json`)

Each cell gives the ratio at w = 0 / 0.1 / 1. The bracket is the 95 percent interval at w = 0, except where a cell carries its own interval.

shape	ratio	w = 0	w = 0.1	w = 1	pooled band order	per curve (lower at every weight)
a3	sib8-p256 / tw2cj-lazy	0.516 [0.494, 0.538]	0.612	1.048	unresolved	1 of 5
a3	sib8-p512 / tw2cj-lazy	0.505 [0.479, 0.532]	0.598	1.020	unresolved	1 of 5
gen	**sib8-p256 / tw2cj-lazy**	**0.486 [0.460, 0.513]**	**0.577**	**0.988**	**sib8-p256 lower at every weight**	4 of 5
gen	sib8-p512 / tw2cj-lazy	0.492 [0.466, 0.519]	0.583	0.995	sib8-p512 lower at every weight	3 of 5
j0b7	sib8-p256 / tw2cj-lazy	0.501 [0.476, 0.527]	0.594	1.017	unresolved	2 of 5
j0b7	**sib8-p512 / tw2cj-lazy**	**0.492 [0.471, 0.515]**	**0.583**	**0.994**	**sib8-p512 lower at every weight**	3 of 5
a3 / gen / j0b7	sib8-p512 / sib8-p256	0.979 / 1.012 / 0.984			p512 lower / higher / lower	
a3 / gen / j0b7	tw2cj-lazy / rho-neg	1.669 / 1.865 / 1.796			rho-neg lower	
a3	best sib / rho-neg	p512: 0.843 [0.786, 0.903]	0.999	1.705	unresolved	unresolved on 5 of 5
gen	best sib / rho-neg	p256: 0.906 [0.848, 0.968]	1.076	1.846	unresolved	rho-neg lower on 1 of 5
j0b7	rho-cubic / rho-neg	0.581 [0.539, 0.626]	0.580	0.577	rho-cubic lower	
j0b7	best sib / rho-cubic	p512: 1.523 [1.416, 1.638]	1.807	3.099	**rho-cubic lower at every weight**	

Bit-ops (bitop/v1, `stats_bitops.py`, `bitops-*.json`):

ratio	a3	gen	j0b7
sib8-p256 / tw2cj-lazy	0.630 [0.598, 0.664]	0.594 [0.563, 0.625]	0.611 [0.582, 0.642]

ratio	a3	gen	j0b7
sib8-p512 / tw2cj-lazy	0.616 [0.585, 0.647]	0.600 [0.568, 0.633]	0.601 [0.571, 0.631]
sib8-p256 / rho-neg	1.023 [0.954, 1.099]	1.077 [1.003, 1.162]	1.069 [0.997, 1.149]
sib8-p512 / rho-neg	1.000 [0.935, 1.068]	1.089 [1.013, 1.171]	1.050 [0.980, 1.126]
rho-cubic / rho-neg			1.095 [1.003, 1.194]


Stage 2
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Against the pre-registered expectations

- **F1 holds.** sib8-p256 / tw2cj-lazy is at most 0.56 on a3 (0.516) and on gen (0.486). The order “lower at every weight” holds pooled on gen.
- **F2 does not hold as stated.** p512 / p256 is 0.979 on a3 and 0.984 on j0b7, but 1.012 on gen, and every interval contains 1. Going from a 256-point to a 512-point pool changes cost by about ±2 percent at this size.
- **F3 holds on a3 and fails on gen.** The bit-op interval of sib8-p256 / rho-neg contains 1 on a3 and j0b7, but lies above 1 on gen: 1.077 [1.003, 1.162]. The band order against rho-neg stays unresolved on every shape.
- **F4 holds in GAE.** rho-cubic / rho-neg is 0.581, the expected sqrt(3) gain. Every IC arm is above rho-cubic at every weight; the best sibling arm is 1.52 to 3.10 times rho-cubic.
- **Open instrument question.** In bit-ops, rho-cubic / rho-neg is 1.095 [1.003, 1.194]: the unit charges rho-cubic more than rho-neg, where GAE says 0.58. The two units disagree on what the order-3 canonicalisation costs. Until that is reconciled, bit-op ratios on j0b7 against rho-cubic are not used here.

What this means for prime-field ECDLP, and what it does not

- **The gain carries over.** Sibling points cut the counted cost of the C = 2 two-seed walk to about half at w = 0 and to 0.6 in bit-ops on all three prime-field shapes, not only on the cofactor-8 Stage 1 family. On gen and j0b7 the gain survives every band weight pooled [pcc:unverified].
- **Distance to rho remains.**

    - Against rho-neg (A = 2), the best sibling arm is level in bit-ops on a3 (1.000 [0.935, 1.068]) and 5 to 9 percent above it on gen and j0b7. It is unresolved in the band everywhere.
    - Against the matched rho for the secp256k1 shape, rho-cubic (A = 6), the distance is 1.5 times at w = 0 and 3.1 times at w = 1. The index-calculus walk has no counterpart to the order-3 automorphism, so that rho has a sqrt(3) head start the walk does not get.
- **Size.** r is at most 2^31, the engine's current limit for `sib` (and 2^37 for `lazy=1, tw=2`). These measurements say nothing about how the ratio scales with p. Lifting the limit is engine work (next steps).
- **Not shown.** No below-rho result. No wall-time result (L0). No EKS run. No leaderboard change. No claim about P-256 or secp256k1 themselves.

Next

1. Engine: widen the `wwalk` coefficient packing so `sib` and `lazy` run at r = 2^34 to 2^40. Stage 3 then measures how the ratio to rho moves with r on these three shapes.
2. Give the walk the order-3 automorphism on j = 0 curves (orbit-canonical keys, as rho-cubic does). Without it, the secp256k1 shape stays more than 1.5 times behind its matched rho.
3. Reconcile the bitop/v1 charge for rho-cubic's canonicalisation against GAE.
4. Validation gate and an EKS L2 re-run once the Iso Lab route works.

Files: `PROTOCOL.md`, `frozen.sha256`, `spec-{a3,gen,j0b7}.json`, `run_local.sh`, `sessions/`, `verify-*.json`, `band-*.{json,txt}`, `results.json`, `collect.py`, `stats_bitops.py`, `bitops-*.json`, `plot.py`, `file:///home/user/crypto/research/curve25519_autolab_20261104/siblings/stage2-primefield/figures/`, `REPORT.pdf`.